Ansible Automation - Use Cases

A reference of the use cases by which Ansible Automation Platform is commonly implemented, by domain. Each use case is described in the same format (Description, Collection, Key content, Notes) and mapped to Red Hat Certified and Validated content.

5 solution use cases · 28 product use cases · generated 2026-10-21 · community reference, not an official Red Hat publication

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Microsoft Azure

Provision and operate Microsoft Azure resources.

Solution use cases

Use Case: Provision Azure networking and virtual machines

Description	Create the base of an Azure environment, a resource group, virtual network, subnets and virtual machines, with a playbook. Environments are built the same way each time, and the same automation can remove them when a project ends.
Collection	`azure.azcollection` (Certified), `cloud.azure_ops` (Validated)
Key content	• `cloud.azure_ops.azure_manage_resource_group` — A role to manage Azure Resource Group. User can create or delete resource group. • `cloud.azure_ops.azure_manage_networking_stack` — A role to Create/Delete/Configure an Azure Network Interface. • `cloud.azure_ops.azure_virtual_machine_with_public_ip` — A role to Create/Delete/Configure an Azure Virtual Machine. • `azure.azcollection.azure_rm_virtualnetwork` — Manage Azure virtual networks • `azure.azcollection.azure_rm_subnet` — Manage Azure subnets • `azure.azcollection.azure_rm_virtualmachine` — Manage Azure virtual machines
Notes	• Add load balancing and network rules with Manage Azure load balancers and security groups. • The `azure_rm_deployment` module in `azure.azcollection` can run existing ARM templates when a template is already the standard. • If Terraform is the standard provisioner, see Run Terraform from Automation Platform.

Use Case: Manage Azure load balancers and security groups

Description	Expose Azure workloads through a load balancer and control access with network security groups, all from version-controlled definitions. Rule changes are requested through a pull request, applied by a job, and visible in history.
Collection	`cloud.azure_ops` (Validated), `azure.azcollection` (Certified)
Key content	• `cloud.azure_ops.azure_load_balancer_with_public_ip` — A role to Create/Delete/Configure an Azure Load Balancer. • `cloud.azure_ops.azure_manage_security_group` — A role to manage an Azure Security Group. User can create or delete a security group, as well as add/remove rules from a security group. • `azure.azcollection.azure_rm_loadbalancer` — Manage Azure load balancers • `azure.azcollection.azure_rm_securitygroup` — Manage Azure network security groups • `azure.azcollection.azure_rm_publicipaddress` — Manage Azure Public IP Addresses • `azure.azcollection.azure_rm_appgateway` — Manage Application Gateway instance

Notes	• Combine with Provision Azure networking and virtual machines for the underlying network. • Pair with Event-Driven Ansible and the `azure.azcollection.azure_event_hub` event source to react to Azure activity log events. • Apply a lock with `azure.azcollection.azure_rm_lock` to protect shared network resources from deletion.

Use Case: Back up and manage Azure resources

Description	Put backup configuration for Azure virtual machines under automation: recovery vaults, policies and VM protection, plus snapshots for changes. Storage accounts and other resources are managed with the same definitions, so protection is not forgotten on new workloads.
Collection	`azure.azcollection` (Certified), `cloud.azure_ops` (Validated)
Key content	• `azure.azcollection.azure_rm_recoveryservicesvault` — Create and Delete Azure Recovery Services vault • `azure.azcollection.azure_rm_backuppolicy` — Manage Azure Backup Policy • `azure.azcollection.azure_rm_backupazurevm` — Back up an Azure Virtual Machine using Azure Backup • `cloud.azure_ops.azure_manage_snapshot` — A role to manage a snapshot to an Azure Virtual Machine. • `azure.azcollection.azure_rm_snapshot` — Manage Azure Snapshot instance • `cloud.azure_ops.azure_manage_storage_account` — A role to manage Azure Storage Accounts.
Notes	• Restore tests should be run regularly into an isolated resource group; do not rely only on the existence of recovery points. • Resource locks, via `azure.azcollection.azure_rm_lock`, protect vaults from accidental deletion. • Related: Provision Azure networking and virtual machines.

Use Case: Use Azure as a dynamic inventory source

Description	Let Azure supply the host list. Virtual machines in one or more subscriptions are synchronized into Automation Platform and grouped by resource group, location or tag, so job templates always target current machines. Servers outside Azure can be registered with Azure Arc to appear in the same view.
Collection	`azure.azcollection` (Certified), `cloud.azure_ops` (Validated)
Key content	• `azure.azcollection.azure_rm` — Azure Resource Manager inventory plugin • `azure.azcollection.azure_kql` — Azure Resource Manager inventory plugin using Graph QL • `azure.azcollection.azure_arc` — A role to Configure an Azure ARC Host. • `cloud.azure_ops.archost_deploy` — Deploy arc agent on arc_hosts • `cloud.azure_ops.azure_configure_archost` — A role to Configure an Azure ARC Host.

Notes	• Use the `azure.azcollection.azure_service_bus` or `azure_event_hub` event source with Event-Driven Ansible to react when a VM is created. • Restrict each service principal to the subscriptions it needs. • See Back up and manage Azure resources for a job that consumes this inventory.

Use Case: Deploy and migrate a web application on Azure

Description	Deliver an application to Azure with a repeatable playbook, check that the deployment works, and move it to another region when needed. Application teams run the workflow as a service; operations teams own the playbooks.
Collection	`cloud.azure_ops` (Validated), `azure.azcollection` (Certified)
Key content	• `cloud.azure_ops.webapp` — Bootstrap a web app application using Azure collection • `cloud.azure_ops.webapp_container` — Deploy existing python app to container instance on Azure • `cloud.azure_ops.validate_deployment` — Validate successful deployment of application • `cloud.azure_ops.vmss_migrate` — Migrate a web app application from one region to another region • `azure.azcollection.azure_rm_webapp` — Manage Web App instances • `azure.azcollection.azure_rm_virtualmachinescaleset` — Manage Azure virtual machine scale sets
Notes	• Pair with Manage Azure load balancers and security groups for traffic management. • `cloud.azure_ops` also includes playbooks for Azure Machine Learning platform tasks, such as `cloud.azure_ops.self_service_ml_platform`. • Keep the old region available until the new one has passed validation and traffic has been switched.

Product use cases

Grouped from the modules of each certified or validated collection. Key content lists the components each use case supports, with the modules that manage them (create, update, delete) and query them (_info).

Microsoft Azure `azure.azcollection`

Certified Microsoft · version 4.1.0 · updated 2026-10-02 · 449 modules → 24 use cases

Use Case: Manage virtual machines and images

Description	Create, resize, start, stop and delete Azure VMs, including extensions, sizes, images, SSH keys and availability constructs.
Collection	`azure.azcollection` (Certified, Microsoft)

Key content	**Virtual machines** `azure_rm_virtualmachine`, `azure_rm_virtualmachine_info` **VM extensions** `azure_rm_virtualmachineextension`, `azure_rm_virtualmachineextension_info` **VM images (marketplace)** `azure_rm_virtualmachineimage_info` **VM sizes** `azure_rm_virtualmachinesize_info` **Compute SKUs** `azure_rm_vmsku_info` **Image SKUs** `azure_rm_imagesku_info` **Custom images** `azure_rm_image`, `azure_rm_image_info` **SSH public keys** `azure_rm_sshpublickey`, `azure_rm_sshpublickey_info` **Availability sets** `azure_rm_availabilityset`, `azure_rm_availabilityset_info` **Proximity placement groups** `azure_rm_proximityplacementgroup`, `azure_rm_proximityplacementgroup_info` **Capacity reservation groups** `azure_rm_capacityreservationgroup`, `azure_rm_capacityreservationgroup_info` **Dedicated hosts** `azure_rm_dedicatedhost`, `azure_rm_dedicatedhost_info` **Dedicated host groups** `azure_rm_hostgroup`, `azure_rm_hostgroup_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage VM scale sets

Description	Create, scale and update virtual machine scale sets, their extensions and individual instances.
Collection	`azure.azcollection` (Certified, Microsoft)
Key content	**VM scale sets** `azure_rm_virtualmachinescaleset`, `azure_rm_virtualmachinescaleset_info` **Scale set extensions** `azure_rm_virtualmachinescalesetextension`, `azure_rm_virtualmachinescalesetextension_info` **Scale set instances** `azure_rm_virtualmachinescalesetinstance`, `azure_rm_virtualmachinescalesetinstance_info` **Scale set network interfaces** `azure_rm_vmssnetworkinterface_info` **Autoscale settings** `azure_rm_autoscale`, `azure_rm_autoscale_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage disks, snapshots and shared image galleries

Description	Provision managed disks, snapshots, disk encryption sets and shared image gallery content.
Collection	`azure.azcollection` (Certified, Microsoft)
Key content	**Managed disks** `azure_rm_manageddisk`, `azure_rm_manageddisk_info` **Multiple managed disks** `azure_rm_multiplemanageddisks` **Snapshots** `azure_rm_snapshot`, `azure_rm_snapshot_info` **Disk access** `azure_rm_diskaccess`, `azure_rm_diskaccess_info` **Disk encryption sets** `azure_rm_diskencryptionset`, `azure_rm_diskencryptionset_info` **Image galleries** `azure_rm_gallery`, `azure_rm_gallery_info` **Gallery images** `azure_rm_galleryimage`, `azure_rm_galleryimage_info` **Gallery image versions** `azure_rm_galleryimageversion`, `azure_rm_galleryimageversion_info`

Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage virtual networks and connectivity

Description	Create virtual networks, subnets, peerings, route tables, NAT gateways and service endpoint policies.
Collection	`azure.azcollection` (Certified, Microsoft)
Key content	**Virtual networks** `azure_rm_virtualnetwork`, `azure_rm_virtualnetwork_info` **Subnets** `azure_rm_subnet`, `azure_rm_subnet_info` **Virtual network peering** `azure_rm_virtualnetworkpeering`, `azure_rm_virtualnetworkpeering_info` **Route tables** `azure_rm_routetable`, `azure_rm_routetable_info` **Routes** `azure_rm_route`, `azure_rm_route_info` **NAT gateways** `azure_rm_natgateway`, `azure_rm_natgateway_info` **Service endpoint policies** `azure_rm_serviceendpointpolicy`, `azure_rm_serviceendpointpolicy_info` **Service endpoint policy definitions** `azure_rm_serviceendpointpolicydefinition`, `azure_rm_serviceendpointpolicydefinition_info` **Network interfaces** `azure_rm_networkinterface`, `azure_rm_networkinterface_info` **Public IP addresses** `azure_rm_publicipaddress`, `azure_rm_publicipaddress_info` **Public IP prefixes** `azure_rm_publicipprefix`, `azure_rm_publicipprefix_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage hybrid, VPN and WAN connectivity

Description	Configure virtual network gateways, VPN sites, ExpressRoute circuits and Virtual WAN hubs.
Collection	`azure.azcollection` (Certified, Microsoft)
Key content	**Virtual network gateways** `azure_rm_virtualnetworkgateway`, `azure_rm_virtualnetworkgateway_info` **Gateway connections** `azure_rm_virtualnetworkgatewayconnection`, `azure_rm_virtualnetworkgatewayconnection_info` **Gateway NAT rules** `azure_rm_virtualnetworkgatewaynatrule`, `azure_rm_virtualnetworkgatewaynatrule_info` **Local network gateways** `azure_rm_localnetworkgateway`, `azure_rm_localnetworkgateway_info` **ExpressRoute circuits** `azure_rm_expressroute`, `azure_rm_expressroute_info` **VPN sites** `azure_rm_vpnsite`, `azure_rm_vpnsite_info` **VPN site links** `azure_rm_vpnsitelink_info` **Virtual WAN** `azure_rm_virtualwan`, `azure_rm_virtualwan_info` **Virtual hubs** `azure_rm_virtualhub`, `azure_rm_virtualhub_info` **Virtual hub connections** `azure_rm_virtualhubconnection`, `azure_rm_virtualhubconnection_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage load balancing and traffic distribution

Description	Configure load balancers, application gateways, Traffic Manager and Front Door profiles.
Collection	`azure.azcollection` (Certified, Microsoft)
Key content	**Load balancers** `azure_rm_loadbalancer`, `azure_rm_loadbalancer_info` **Backend address pools** `azure_rm_loadbalancerbackendaddresspool`, `azure_rm_loadbalancerbackendaddresspool_info` **Application gateways** `azure_rm_appgateway`, `azure_rm_appgateway_info` **Traffic Manager** `azure_rm_trafficmanager` **Traffic Manager profiles** `azure_rm_trafficmanagerprofile`, `azure_rm_trafficmanagerprofile_info` **Traffic Manager endpoints** `azure_rm_trafficmanagerendpoint`, `azure_rm_trafficmanagerendpoint_info` **Front Door endpoints** `azure_rm_afdendpoint`, `azure_rm_afdendpoint_info` **Front Door origins** `azure_rm_afdorigin`, `azure_rm_afdorigin_info` **Front Door origin groups** `azure_rm_afdorigingroup`, `azure_rm_afdorigingroup_info` **Front Door routes** `azure_rm_afdroute`, `azure_rm_afdroute_info` **Front Door rules** `azure_rm_afdrules`, `azure_rm_afdrules_info` **Front Door rule sets** `azure_rm_afdruleset`, `azure_rm_afdruleset_info` **CDN profiles** `azure_rm_cdnprofile`, `azure_rm_cdnprofile_info` **CDN endpoints** `azure_rm_cdnendpoint`, `azure_rm_cdnendpoint_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage network security and firewalls

Description	Configure network security groups, application security groups, Azure Firewall, firewall policies, WAF policies, DDoS plans and Bastion.
Collection	`azure.azcollection` (Certified, Microsoft)
Key content	**Network security groups** `azure_rm_securitygroup`, `azure_rm_securitygroup_info` **Application security groups** `azure_rm_applicationsecuritygroup`, `azure_rm_applicationsecuritygroup_info` **Azure Firewall** `azure_rm_azurefirewall`, `azure_rm_azurefirewall_info` **Firewall policies** `azure_rm_firewallpolicy`, `azure_rm_firewallpolicy_info` **Firewall rule collection groups** `azure_rm_firewallpolicyrulecollectiongroup`, `azure_rm_firewallpolicyrulecollectiongroup_info` **IP groups** `azure_rm_ipgroup`, `azure_rm_ipgroup_info` **Web application firewall policies** `azure_rm_applicationfirewallpolicy`, `azure_rm_applicationfirewallpolicy_info` **DDoS protection plans** `azure_rm_ddosprotectionplan`, `azure_rm_ddosprotectionplan_info` **Bastion hosts** `azure_rm_bastionhost`, `azure_rm_bastionhost_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage private connectivity and DNS

Description	Create private endpoints, private link services and public or private DNS zones and records.

Collection	`azure.azcollection` (Certified, Microsoft)
Key content	**Private endpoints** `azure_rm_privateendpoint`, `azure_rm_privateendpoint_info` **Private endpoint connections** `azure_rm_privateendpointconnection`, `azure_rm_privateendpointconnection_info` **Private endpoint DNS zone groups** `azure_rm_privateendpointdnszonegroup`, `azure_rm_privateendpointdnszonegroup_info` **Private link services** `azure_rm_privatelinkservice`, `azure_rm_privatelinkservice_info` **DNS zones** `azure_rm_dnszone`, `azure_rm_dnszone_info` **DNS record sets** `azure_rm_dnsrecordset`, `azure_rm_dnsrecordset_info` **Private DNS zones** `azure_rm_privatednszone`, `azure_rm_privatednszone_info` **Private DNS record sets** `azure_rm_privatednsrecordset`, `azure_rm_privatednsrecordset_info` **Private DNS zone links** `azure_rm_privatednszonelink`, `azure_rm_privatednszonelink_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage Kubernetes and OpenShift clusters

Description	Create and update AKS clusters and node pools, query available versions and credentials, and manage Azure Red Hat OpenShift clusters.
Collection	`azure.azcollection` (Certified, Microsoft)
Key content	**AKS clusters** `azure_rm_aks`, `azure_rm_aks_info` **AKS node pools** `azure_rm_aksagentpool`, `azure_rm_aksagentpool_info` **Node pool versions** `azure_rm_aksagentpoolversion_info` **AKS credentials** `azure_rm_akscredentials_info` **AKS upgrade versions** `azure_rm_aksupgrade_info` **Kubernetes versions** `azure_rm_aksversion_info` **OpenShift clusters** `azure_rm_openshiftmanagedcluster`, `azure_rm_openshiftmanagedcluster_info` **OpenShift kubeconfig** `azure_rm_openshiftmanagedclusterkubeconfig_info` **OpenShift versions** `azure_rm_openshiftmanagedclusterversion_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage containers and container registries

Description	Run container instances and Container Apps, and manage registries with their replications, tokens, scope maps, tags and webhooks.
Collection	`azure.azcollection` (Certified, Microsoft)

Key content	**Container instances** `azure_rm_containerinstance`, `azure_rm_containerinstance_info` **Container Apps** `azure_rm_containerapp`, `azure_rm_containerapp_info` **Container Apps environments** `azure_rm_containerappenvironment`, `azure_rm_containerappenvironment_info` **Container registries** `azure_rm_containerregistry`, `azure_rm_containerregistry_info` **Registry replications** `azure_rm_containerregistryreplication`, `azure_rm_containerregistryreplication_info` **Registry scope maps** `azure_rm_containerregistryscopemap`, `azure_rm_containerregistryscopemap_info` **Registry tags** `azure_rm_containerregistrytag`, `azure_rm_containerregistrytag_info` **Registry tokens** `azure_rm_containerregistrytoken`, `azure_rm_containerregistrytoken_info` **Registry token passwords** `azure_rm_containerregistrytokenpassword` **Registry webhooks** `azure_rm_containerregistrywebhook`, `azure_rm_containerregistrywebhook_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage web apps and serverless

Description	Deploy and configure App Service plans, web apps, deployment slots, Function Apps and API Management.
Collection	`azure.azcollection` (Certified, Microsoft)
Key content	**App Service plans** `azure_rm_appserviceplan`, `azure_rm_appserviceplan_info` **Web apps** `azure_rm_webapp`, `azure_rm_webapp_info` **Web app slots** `azure_rm_webappslot` **Web app access restrictions** `azure_rm_webappaccessrestriction`, `azure_rm_webappaccessrestriction_info` **Web app virtual network connections** `azure_rm_webappvnetconnection`, `azure_rm_webappvnetconnection_info` **Function Apps** `azure_rm_functionapp`, `azure_rm_functionapp_info` **API Management APIs** `azure_rm_apimanagement`, `azure_rm_apimanagement_info` **API Management services** `azure_rm_apimanagementservice`, `azure_rm_apimanagementservice_info` **App Configuration stores** `azure_rm_appconfiguration`, `azure_rm_appconfiguration_info` **App Configuration key-values** `azure_rm_appconfigurationkey`, `azure_rm_appconfigurationkey_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage storage

Description	Create storage accounts, blob containers, file shares and lifecycle management policies.
Collection	`azure.azcollection` (Certified, Microsoft)

Key content	**Storage accounts** `azure_rm_storageaccount`, `azure_rm_storageaccount_info` **Storage management policies** `azure_rm_storageaccountmanagementpolicy`, `azure_rm_storageaccountmanagementpolicy_info` **Blob containers and objects** `azure_rm_storageblob`, `azure_rm_storageblob_info` **File shares** `azure_rm_storageshare`, `azure_rm_storageshare_info` **File share directories** `azure_rm_storagesharedirectory`, `azure_rm_storagesharedirectory_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage SQL databases

Description	Manage Azure SQL servers, databases, elastic pools, firewall rules and managed instances including retention policies.
Collection	`azure.azcollection` (Certified, Microsoft)
Key content	**SQL servers** `azure_rm_sqlserver`, `azure_rm_sqlserver_info` **SQL databases** `azure_rm_sqldatabase`, `azure_rm_sqldatabase_info` **Elastic pools** `azure_rm_sqlelasticpool`, `azure_rm_sqlelasticpool_info` **SQL firewall rules** `azure_rm_sqlfirewallrule`, `azure_rm_sqlfirewallrule_info` **SQL managed instances** `azure_rm_sqlmanagedinstance`, `azure_rm_sqlmanagedinstance_info` **Managed instance databases** `azure_rm_sqlmidatabase`, `azure_rm_sqlmidatabase_info` **Long-term retention policies** `azure_rm_sqlmidblongtermretentionpolicy`, `azure_rm_sqlmidblongtermretentionpolicy_info` **Short-term retention policies** `azure_rm_sqlmidbshorttermretentionpolicy`, `azure_rm_sqlmidbshorttermretentionpolicy_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage open source and NoSQL databases and caches

Description	Manage MySQL and PostgreSQL flexible servers, Cosmos DB accounts and Azure Cache for Redis.
Collection	`azure.azcollection` (Certified, Microsoft)

Key content	**MySQL flexible servers** `azure_rm_mysqlflexibleserver`, `azure_rm_mysqlflexibleserver_info` **MySQL databases** `azure_rm_mysqlflexibledatabase`, `azure_rm_mysqlflexibledatabase_info` **MySQL configurations** `azure_rm_mysqlflexibleconfiguration`, `azure_rm_mysqlflexibleconfiguration_info` **MySQL firewall rules** `azure_rm_mysqlflexiblefirewallrule`, `azure_rm_mysqlflexiblefirewallrule_info` **PostgreSQL flexible servers** `azure_rm_postgresqlflexibleserver`, `azure_rm_postgresqlflexibleserver_info` **PostgreSQL databases** `azure_rm_postgresqlflexibledatabase`, `azure_rm_postgresqlflexibledatabase_info` **PostgreSQL configurations** `azure_rm_postgresqlflexibleconfiguration`, `azure_rm_postgresqlflexibleconfiguration_info` **PostgreSQL firewall rules** `azure_rm_postgresqlflexiblefirewallrule`, `azure_rm_postgresqlflexiblefirewallrule_info` **PostgreSQL administrators** `azure_rm_postgresqlflexibleadministrator`, `azure_rm_postgresqlflexibleadministrator_info` **PostgreSQL backups** `azure_rm_postgresqlflexiblebackup`, `azure_rm_postgresqlflexiblebackup_info` **PostgreSQL virtual endpoints** `azure_rm_postgresqlflexiblevirtualendpoint`, `azure_rm_postgresqlflexiblevirtualendpoint_info` **Cosmos DB accounts** `azure_rm_cosmosdbaccount`, `azure_rm_cosmosdbaccount_info` **Redis caches** `azure_rm_rediscache`, `azure_rm_rediscache_info` **Redis firewall rules** `azure_rm_rediscachefirewallrule`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage identity and directory objects

Description	Manage Microsoft Entra ID (Azure AD) applications, groups, users, service principals, passwords, managed identities and federated credentials.
Collection	`azure.azcollection` (Certified, Microsoft)
Key content	**Entra ID applications** `azure_rm_adapplication`, `azure_rm_adapplication_info` **Entra ID groups** `azure_rm_adgroup`, `azure_rm_adgroup_info` **Application passwords** `azure_rm_adpassword`, `azure_rm_adpassword_info` **Service principals** `azure_rm_adserviceprincipal`, `azure_rm_adserviceprincipal_info` **Entra ID users** `azure_rm_aduser`, `azure_rm_aduser_info` **User-assigned managed identities** `azure_rm_userassignedidentity`, `azure_rm_userassignedidentity_info` **Federated identity credentials** `azure_rm_federatedidentitycredential`, `azure_rm_federatedidentitycredential_info` **API access tokens** `azure_rm_accesstoken_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage access control and governance

Description	Manage role definitions and assignments, resource locks, tags, management groups, resource groups, subscriptions and delegated management.

Collection	`azure.azcollection` (Certified, Microsoft)
Key content	**Role definitions** `azure_rm_roledefinition`, `azure_rm_roledefinition_info` **Role assignments** `azure_rm_roleassignment`, `azure_rm_roleassignment_info` **Resource locks** `azure_rm_lock`, `azure_rm_lock_info` **Tags** `azure_rm_tags`, `azure_rm_tags_info` **Management groups** `azure_rm_managementgroup`, `azure_rm_managementgroup_info` **Resource groups** `azure_rm_resourcegroup`, `azure_rm_resourcegroup_info` **Subscriptions** `azure_rm_subscription_info` **Account details** `azure_rm_account_info` **Lighthouse registration assignments** `azure_rm_registrationassignment`, `azure_rm_registrationassignment_info` **Lighthouse registration definitions** `azure_rm_registrationdefinition`, `azure_rm_registrationdefinition_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage secrets, keys and certificates

Description	Manage Key Vault instances and their secrets, keys, certificates and HSM security domains, and read secrets in playbooks.
Collection	`azure.azcollection` (Certified, Microsoft)
Key content	**Key vaults** `azure_rm_keyvault`, `azure_rm_keyvault_info` **Key Vault secrets** `azure_rm_keyvaultsecret`, `azure_rm_keyvaultsecret_info` **Key Vault keys** `azure_rm_keyvaultkey`, `azure_rm_keyvaultkey_info` **Key Vault certificates** `azure_rm_keyvaultcertificate`, `azure_rm_keyvaultcertificate_info` **Key Vault security domains** `azure_rm_keyvaultsecuritydomain` **Key Vault secret lookup** lookup `azure_keyvault_secret` **Service principal attribute lookup** lookup `azure_service_principal_attribute`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Monitor and alert on Azure resources

Description	Configure Log Analytics workspaces, diagnostic settings, metric and activity log alerts, action groups, data collection rules, network flow logs and Network Watcher.
Collection	`azure.azcollection` (Certified, Microsoft)

Key content	**Log Analytics workspaces** `azure_rm_loganalyticsworkspace`, `azure_rm_loganalyticsworkspace_info` **Diagnostic settings** `azure_rm_monitordiagnosticsetting`, `azure_rm_monitordiagnosticsetting_info` **Log profiles** `azure_rm_monitorlogprofile` **Metric alerts** `azure_rm_monitormetricalerts`, `azure_rm_monitormetricalerts_info` **Activity log alerts** `azure_rm_monitoractivitylogalerts`, `azure_rm_monitoractivitylogalerts_info` **Scheduled query alerts** `azure_rm_monitorscheduledqueryrules`, `azure_rm_monitorscheduledqueryrules_info` **Action groups** `azure_rm_monitoractiongroups`, `azure_rm_monitoractiongroups_info` **Data collection endpoints** `azure_rm_monitordatacollectionendpoint`, `azure_rm_monitordatacollectionendpoint_info` **Data collection rules** `azure_rm_monitordatacollectionrules`, `azure_rm_monitordatacollectionrules_info` **Data collection rule associations** `azure_rm_monitordatacollectionruleassociation`, `azure_rm_monitordatacollectionruleassociation_info` **Network flow logs** `azure_rm_networkflowlog`, `azure_rm_networkflowlog_info` **Network Watcher** `azure_rm_networkwatcher`, `azure_rm_networkwatcher_info` **Resource health states** `azure_rm_resourcehealthstates_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Protect and back up workloads

Description	Configure Recovery Services vaults, backup vaults, backup policies and VM or disk backups, and check backup jobs.
Collection	`azure.azcollection` (Certified, Microsoft)
Key content	**Recovery Services vaults** `azure_rm_recoveryservicesvault`, `azure_rm_recoveryservicesvault_info` **Recovery Services vault configuration** `azure_rm_recoveryservicesvaultconfig`, `azure_rm_recoveryservicesvaultconfig_info` **VM backups** `azure_rm_backupazurevm`, `azure_rm_backupazurevm_info` **Backup policies** `azure_rm_backuppolicy`, `azure_rm_backuppolicy_info` **VM backup policies** `azure_rm_vmbackuppolicy`, `azure_rm_vmbackuppolicy_info` **Backup jobs** `azure_rm_backupjob_info` **Backup vaults** `azure_rm_dataprotectionbackupvault`, `azure_rm_dataprotectionbackupvault_info` **Disk backup instances** `azure_rm_dataprotectionbackupinstancedisk`, `azure_rm_dataprotectionbackupinstancedisk_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Deploy infrastructure from templates and generic resources

Description	Deploy ARM template deployments or create any Azure resource through the generic resource module.
Collection	`azure.azcollection` (Certified, Microsoft)

Key content	**ARM template deployments** `azure_rm_deployment`, `azure_rm_deployment_info` **Generic Azure resources** `azure_rm_resource`, `azure_rm_resource_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage data, analytics and AI services

Description	Manage Azure Machine Learning workspaces, compute, models, jobs and endpoints; AI Search; Cognitive Services; Databricks, Data Factory and HDInsight; and Batch accounts.
Collection	`azure.azcollection` (Certified, Microsoft)
Key content	**ML workspaces** `azure_rm_ml_workspace`, `azure_rm_ml_workspace_info` **ML registries** `azure_rm_ml_registry`, `azure_rm_ml_registry_info` **ML compute** `azure_rm_ml_compute`, `azure_rm_ml_compute_info` **ML connections** `azure_rm_ml_connection`, `azure_rm_ml_connection_info` **ML data assets** `azure_rm_ml_data`, `azure_rm_ml_data_info` **ML datastores** `azure_rm_ml_datastore`, `azure_rm_ml_datastore_info` **ML environments** `azure_rm_ml_environment`, `azure_rm_ml_environment_info` **ML components** `azure_rm_ml_component`, `azure_rm_ml_component_info` **ML models** `azure_rm_ml_model`, `azure_rm_ml_model_info` **ML jobs** `azure_rm_ml_job`, `azure_rm_ml_job_info` **ML schedules** `azure_rm_ml_schedule`, `azure_rm_ml_schedule_info` **ML batch endpoints** `azure_rm_ml_batch_endpoint`, `azure_rm_ml_batch_endpoint_info` **ML batch deployments** `azure_rm_ml_batch_deployment`, `azure_rm_ml_batch_deployment_info` **ML batch endpoint invocation** `azure_rm_ml_batch_endpoint_invoke` **ML online endpoints** `azure_rm_ml_online_endpoint`, `azure_rm_ml_online_endpoint_info` **ML online deployments** `azure_rm_ml_online_deployment`, `azure_rm_ml_online_deployment_info` **ML online endpoint invocation** `azure_rm_ml_online_endpoint_invoke` **AI Search services** `azure_rm_cognitivesearch`, `azure_rm_cognitivesearch_info` **Search data sources** `azure_rm_searchdatasource`, `azure_rm_searchdatasource_info` **Search indexes** `azure_rm_searchindex`, `azure_rm_searchindex_info` **Search indexers** `azure_rm_searchindexer`, `azure_rm_searchindexer_info` **Search skillsets** `azure_rm_searchskillset`, `azure_rm_searchskillset_info` **Search synonym maps** `azure_rm_searchsynonymmap`, `azure_rm_searchsynonymmap_info` **Cognitive Services accounts** `azure_rm_cognitiveservicesaccount`, `azure_rm_cognitiveservicesaccount_info` **Model deployments** `azure_rm_cognitiveservicesdeployment`, `azure_rm_cognitiveservicesdeployment_info` **Available models** `azure_rm_cognitiveservicesmodel_info` **Databricks workspaces** `azure_rm_databricksworkspace`, `azure_rm_databricksworkspace_info` **Data factories** `azure_rm_datafactory`, `azure_rm_datafactory_info` **HDInsight clusters** `azure_rm_hdinsightcluster`, `azure_rm_hdinsightcluster_info` **Batch accounts** `azure_rm_batchaccount`, `azure_rm_batchaccount_info` **Batch applications** `azure_rm_batchaccountapplication`, `azure_rm_batchaccountapplication_info` **Batch application packages** `azure_rm_batchaccountapplicationpackage`, `azure_rm_batchaccountapplicationpackage_info` **Batch pools** `azure_rm_batchaccountpool`, `azure_rm_batchaccountpool_info`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Manage messaging, events and IoT

Description	Manage Event Hubs, Event Grid topics and subscriptions, Service Bus, Notification Hubs and IoT Hub, and consume Event Hub and Service Bus events in Event-Driven Ansible.
Collection	`azure.azcollection` (Certified, Microsoft)
Key content	**Event Hubs** `azure_rm_eventhub`, `azure_rm_eventhub_info` **Event Grid topics** `azure_rm_eventgrid_topic`, `azure_rm_eventgrid_topic_info` **Event Grid subscriptions** `azure_rm_eventgrid_subscription`, `azure_rm_eventgrid_subscription_info` **Event Grid topic subscriptions** `azure_rm_eventgrid_topic_subscription` **Service Bus namespaces** `azure_rm_servicebus`, `azure_rm_servicebus_info` **Service Bus queues** `azure_rm_servicebusqueue` **Service Bus topics** `azure_rm_servicebustopic` **Service Bus subscriptions** `azure_rm_servicebustopicsubscription` **Service Bus rules** `azure_rm_servicebusrule`, `azure_rm_servicebusrule_info` **Service Bus SAS policies** `azure_rm_servicebussaspolicy` **Notification Hubs** `azure_rm_notificationhub`, `azure_rm_notificationhub_info` **IoT hubs** `azure_rm_iothub`, `azure_rm_iothub_info` **IoT Hub consumer groups** `azure_rm_iothubconsumergroup` **IoT devices** `azure_rm_iotdevice`, `azure_rm_iotdevice_info` **IoT device modules** `azure_rm_iotdevicemodule` **Event Hub event source** eda event source `azure_event_hub` **Service Bus event source** eda event source `azure_service_bus`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Automate operations and DevTest environments

Description	Manage Azure Automation accounts and runbooks and DevTest Labs including VMs, networks, images, policies and schedules.
Collection	`azure.azcollection` (Certified, Microsoft)
Key content	**Automation accounts** `azure_rm_automationaccount`, `azure_rm_automationaccount_info` **Automation runbooks** `azure_rm_automationrunbook`, `azure_rm_automationrunbook_info` **DevTest Labs** `azure_rm_devtestlab`, `azure_rm_devtestlab_info` **Lab ARM templates** `azure_rm_devtestlabarmtemplate_info` **Lab artifacts** `azure_rm_devtestlabartifact_info` **Lab artifact sources** `azure_rm_devtestlabartifactsource`, `azure_rm_devtestlabartifactsource_info` **Lab custom images** `azure_rm_devtestlabcustomimage`, `azure_rm_devtestlabcustomimage_info` **Lab environments** `azure_rm_devtestlabenvironment`, `azure_rm_devtestlabenvironment_info` **Lab policies** `azure_rm_devtestlabpolicy`, `azure_rm_devtestlabpolicy_info` **Lab schedules** `azure_rm_devtestlabschedule`, `azure_rm_devtestlabschedule_info` **Lab virtual machines** `azure_rm_devtestlabvirtualmachine`, `azure_rm_devtestlabvirtualmachine_info` **Lab virtual networks** `azure_rm_devtestlabvirtualnetwork`, `azure_rm_devtestlabvirtualnetwork_info`

Notes	• Most modules cover both management and read-only info queries for the same resource.

Use Case: Discover Azure inventory and onboard Arc servers

Description	Build dynamic inventory from Azure Resource Manager or Resource Graph queries, onboard Azure Arc hosts and manage Arc extensions and SSH access.
Collection	`azure.azcollection` (Certified, Microsoft)
Key content	**Azure Resource Manager inventory** inventory `azure_rm` **Resource Graph (KQL) inventory** inventory `azure_kql` **Azure Arc host configuration** role `azure_arc` **Arc machine extensions** `azure_rm_arcmachineextensions`, `azure_rm_arcmachineextensions_info` **Arc SSH proxy** `azure_rm_arcssh`
Notes	• Most modules cover both management and read-only info queries for the same resource.

Microsoft Azure `cloud.azure_ops`

Validated Red Hat · version 6.2.0 · updated 2026-09-30 · 0 modules → 4 use cases

Use Case: Provision Azure infrastructure

Description	Create and configure resource groups, virtual machines, networking, load balancers, security groups, snapshots, storage accounts and PostgreSQL flexible servers.
Collection	`cloud.azure_ops` (Validated, Red Hat)
Key content	**Resource groups** role `azure_manage_resource_group` **Virtual machines** role `azure_virtual_machine_with_public_ip` **Networking stack** role `azure_manage_networking_stack` **Network interfaces** role `azure_manage_network_interface` **Load balancers** role `azure_load_balancer_with_public_ip` **Security groups** role `azure_manage_security_group` **VM snapshots** role `azure_manage_snapshot` **Storage accounts** role `azure_manage_storage_account` **PostgreSQL flexible servers** role `azure_manage_postgresqlflex` **Microsoft package repository** role `azure_repo_microsoft_prod`

Use Case: Deploy and migrate web applications

Description	Bootstrap web apps and container apps, validate deployments and migrate applications between regions.
Collection	`cloud.azure_ops` (Validated, Red Hat)

Key content	**Web app** playbook `webapp` **Container instance app** playbook `webapp_container` **Cross-region migration** playbook `vmss_migrate` **Deployment validation** playbook `validate_deployment`

Use Case: Run machine learning and AI platforms

Description	Manage Azure AI Search and Cognitive Services accounts, automate MLOps lifecycles, deploy across environments with promotion gates and optimize ML costs.
Collection	`cloud.azure_ops` (Validated, Red Hat)
Key content	**AI Search** role `azure_manage_ai_search` **Cognitive Services accounts** role `azure_manage_cognitive_account` **Self-service ML platform** role `self_service_ml_platform` **Multi-environment ML deployment** role `multi_env_ml_deployment` **ML cost optimization** role `azure_ml_cost_optimization` **Self-service ML platform playbook** playbook `self_service_ml_platform` **Multi-environment ML playbook** playbook `multi_env_ml_deployment` **MLOps lifecycle** playbook `mlops_lifecycle` **RAG pipeline** playbook `rag_pipeline` **ML and AI cost optimization** playbook `cost_optimization`

Use Case: Onboard Azure Arc hosts

Description	Configure Arc hosts, deploy the Arc agent and set up the Arc SSH proxy.
Collection	`cloud.azure_ops` (Validated, Red Hat)
Key content	**Arc host configuration** role `azure_configure_archost` **Arc agent deployment** playbook `archost_deploy` **Arc SSH proxy** playbook `archost_proxy`